Magzhan Ziyabek

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Education

Daegu Gyeongbuk Institute of Science and Technology (DGIST) — Daegu, KR
B.S. in Computer Science and B.E. in Electronics Engineering (English-taught) | GPA: 3.84/4.3 — *Aug. 2024 to Present*

Stanford University — Stanford, CA, USA
Visiting Summer Student | Data Structures; Financial Decision-Making — *Summer 2025*

Jambyl Innovation High College — Taraz, KZ
Diploma in Mathematics Education (English-taught) | GPA: 4.0/4.0 — *Sept. 2020 to July 2024*

Research Experience

Soft Augmented Intelligence Lab (SAIL), DGIST — Daegu, KR
Undergraduate Researcher | Grasp-Intention Inference for Bimanual Soft Gloves — *Aug. 2026 to Present*

- Built a head-mounted RGB-D pipeline that infers from motion alone when each hand of a soft-glove wearer reaches to grasp: 3-D hand points, SAM 2 objects over a table plane, offline grasp proposals.
- Defined the evaluation (first-proposal timing vs. reach onsets, unwanted-proposal burden) and fine-tuned the hand detector for gloves: 92 of 96 gloved hands found on an untouched clip (0 before).

Summer Research Intern | Lab Automation Systems — *Jul. 2026*

- Built and deployed a multi-agent automation system used daily by an eight-person lab. Researchers submit requests through Slack or Notion; the system runs scheduled jobs, health checks and recovery, versioned backups, and audit logs.
- Automated procurement and documentation tasks, including Korean-to-English wiki mirroring, document summaries, vendor and part research, receipt OCR, and purchase paperwork. Cut 2 to 3 hours of manual work per purchase down to one review step.

AI-Vision Lab, DGIST Research Institute — Daegu, KR
AI Star Fellow (Mar. to Jun.); Research Intern (Jan.) | Medical Anomaly Detection — *Jan. 2026 to Jun. 2026*

- Implemented THOR, a diffusion-based anomaly detection pipeline, on 450+ retinal fundus images from Seoul Hospital. Generated pixel-level anomaly maps for unsupervised lesion localization.
- Added field-of-view masking and morphological erosion to reduce false positives caused by imaging artifacts at the retinal boundary.
- Achieved an AUROC of 0.81 for image-level detection of diabetic retinopathy versus normal images, without lesion annotations.

Undergraduate Group Research Program (UGRP), DGIST — Daegu, KR
Undergraduate Researcher | Continual Learning — *Jan. 2026 to Present*

- Received KRW 7M in competitive funding for a study of catastrophic forgetting in CNNs and Vision Transformers trained on sequential tasks.

Teaching Experience

DGIST, School of Undergraduate Studies — Daegu, KR
Undergraduate Teaching Assistant — *Aug. 2025 to Present*

- *Introduction to Programming* (Aug. 2026 to Present): run Python lab sessions and help students work through programming exercises.
- *Single Variable Calculus* (Aug. 2025 to Jan. 2026): graded weekly assignments and quizzes, held office hours, and led MATLAB recitation sessions.

Secondary Mathematics Instruction — Taraz, KZ
Teaching Intern and Mathematics Olympiad Coach — *Apr. 2023 to Mar. 2024*

- Taught algebra and geometry to students in grades 7 to 9; coached students for Olympiad problems in combinatorics, geometry, and algebra.
- Co-produced a 69-video, open-access YouTube series for grades 7 to 9. The series has 14,400+ views.

Projects

LineageGuard | TypeScript, npm

- Open-source reliability layer for multi-agent AI systems that detects where a claim's evidence, meaning, or authority first changes across agent handoffs; zero runtime dependencies, published on npm.

Memory Encryption Engine Performance Modeling | C++, Macsim

- Implemented three memory security schemes in the Macsim architectural simulator (counter-mode encryption, per-cache-line MAC authentication, and Bonsai Merkle Tree replay protection), modeling metadata DRAM requests and critical-path latency across 16 GiB of physical memory.
- Validated on a 4-core x86 out-of-order configuration, reproducing the expected 2×, 3×, and 10× DRAM traffic ratios with 1.51%, 3.28%, and 7.06% execution-time overhead.

Two-Stage In-Order RISC Processor | Verilog, ModelSim

- Designed a 16-bit two-stage (fetch/execute) RISC CPU with an 8-register file, CLA-based ALU, 4-stage bidirectional barrel shifter, and 5-bit FLCNZ status register.
- Supported arithmetic, logic, shift, load/store, branch, jump, and JAL instructions with 16 branch conditions and NOP injection on taken branches; integrated and verified seven modules through full-program simulation.

Hardware Calculator | VHDL, Vivado, Artix-7 FPGA

- Built a 4-bit signed (two's-complement) calculator for addition, subtraction, multiplication, and division without external arithmetic libraries.
- Implemented iterative FSM-based multiplier and divider units with a start/done handshake and Euclidean sign correction, plus a seven-segment display driver; verified by testbench and synthesized for an Artix-7 board.

CS:APP Systems Labs | C, x86-64 Assembly, GDB

- *Bomb Lab* and *Attack Lab:* reverse-engineered x86-64 binaries in GDB, and exploited stack buffer overflows via code injection and return-oriented programming.
- *Malloc*, *Shell* and *Cache Labs:* implemented a custom allocator with free-list design, a Unix shell with job control and signal handling, and an LRU cache simulator with a locality-optimized matrix transpose.

Multi-Bank ATM System | C++

- Designed a modular, object-oriented banking system using finite state machines, supporting multiple banks, account types, transaction history, and a bilingual command-line interface (group project).

Sikgu | Shared food-order prototype | AI Build Week Hackathon Jul. 2026

- Built a prototype for creating shared order rooms and coordinating participants. Kept delivery and payment outside the app to avoid handling regulated transactions.

Awards and Honors

AI Star Fellowship, DGIST AI-Vision Lab Daegu, KR
Competitive undergraduate research fellowship *Mar. 2026 to Jun. 2026*

Dean's List, DGIST Daegu, KR
Semester GPA 4.25/4.3 (threshold ≥ 4.0) *Fall 2024*

National Stand-up Talent Tournament, Kazakhstan Astana, KZ
1st place as team lead and scriptwriter; 1 of 10 finalist teams, performed for 1,200 attendees *May 2024*

Leadership and Outreach

DGIST International Student Ambassador Daegu, KR
Ambassador *Sept. 2024 to Dec. 2024*

- Ran 6 admissions webinars (10 hours total) and answered questions from prospective international applicants in a group chat with 210+ members.